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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/300,359	04/13/2023	Yi-Hau Shiau	129006-US-PA	4251

31561 7590 02/29/2024

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EXAMINER	
KHAN, IBRAHIM A	

ART UNIT	PAPER NUMBER
2619	

NOTIFICATION DATE	DELIVERY MODE
02/29/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/300,359	Applicant(s) Shiau, Yi-Hau	
	Examiner IBRAHIM A KHAN	Art Unit 2619	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 04/13/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,4,6-7,9-14 and 18-20 is/are rejected.

8) ☑ Claim(s) 3,5,8 and 15-17 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☑ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

In the response to this office action, the Examiner respectfully requests that support be shown for language added to any original claims on amendment and any new claims. That is, indicate support for newly added claim language by specifically pointing to page(s) and line numbers in the specification and/or drawing figure(s). This will assist the Examiner in prosecuting this application.

I. ALLOWABLE SUBJECT MATTER

Claims 3, 5, 8, 15-17, objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims. Claim 3 and 15 are objected to because the cited references do not disclose "wherein the functional interface layer comprises at least one solar element, and the at least one solar element covers the ultrasonic fingerprint identification assembly. Claims 5 and 16 are objected to because the cited references do not disclose "wherein the shield layer has a plurality of air bubbles, and each of the air bubbles has a size of less than 5 microns". Claims 8 and 17 are objected to because the cited references do not disclose "wherein the gap is greater than 0.15 mm".

II. INFORMATION DISCLOSURE STATEMENT

The information disclosure statement filed 04/13/2023, has been acknowledged and considered by the examiner. An initialed copy of the PTO-1449 is included in this correspondence.

III. CLAIM REJECTIONS - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 , if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

1. Claims 1-2, 4, 6, 9, and 13-14 are rejected under 35 U.S.C. 103 as being unpatentable over Wang US 20190248439 in view of Panchawagh et al. US 20200160018.

Consider claim 1. Wang discloses a **dashboard device** *[0026] [087] fig 4 user interface comprise a dashboard 148*, **comprising:**

a device body *fig 4 display is housed in body that attaches to handlebar;*

a functional interface layer disposed on the device body *fig 4 [0087]LED tactile display;* **and**

a fingerprint identification assembly disposed in the device and covered by the functional interface layer *[0107-0108] at surface of anti-glare screen 302 is image of fingerprint 336 where finger print is scanned. fingerprint 336 acquisition. [0135] the LED tactile display **148** screen provides a biometric reader for acquiring a fingerprint. [0152] sensor acquires fingerprint from the led tactile display.*

Wang does not disclose **ultrasonic fingerprint identification assembly on the device body**

Panchawagh however discloses **ultrasonic fingerprint identification assembly** *[0030] ultrasonic biometric sensor [0044] receive ultrasonic sensor signal corresponding to finger [0055] fingerprint sensor* **on the device body** *see fig 1A-B device 105 [0038] device 105 maybe auto displays including odometer and speedometer displays fig 2. [0039] the sensor 110 may be attached to or mounted on a frame of the device **105**.*

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the dashboard device of Wang to include **ultrasonic fingerprint identification assembly on the device body**, as taught by Panchawagh, to enable fingerprint reading in a wider lighting and temperature ranges.

Consider claim 2. Wang as modified by Panchawagh disclose **the dashboard device according to claim 1, wherein the functional interface layer comprises a display panel** *Wang fig 4 148*, **and the display panel covers the ultrasonic fingerprint identification assembly** *Panchawagh fig 2 display 220cpvers sensor 202.*

Motivation to combine is similar to motivation in claim 1.

Consider claim 4. Wang as modified by Panchawagh disclose **the dashboard device according to claim 1, wherein the functional interface layer comprises a shield layer, and the shield layer covers the ultrasonic fingerprint identification assembly** *Panchawagh fig 2 cover glass covers the sensor 202.*

Motivation to combine is similar to motivation in claim 1.

Consider claim 6. Wang as modified by Panchawagh disclose **the dashboard device according to claim 1, comprising a double-sided adhesive, wherein the ultrasonic fingerprint identification assembly is glued to the functional interface layer through the double-sided adhesive** *Panchawagh fig 2 adhesive 225.*

Consider claim 9. Wang as modified by Panchawagh disclose **the dashboard device according to claim 1, comprising a light-transmitting protection plate, wherein the light-transmitting protection plate covers the functional interface layer** *Panchawagh fig 2 cover glass covers the sensor 202.*

Motivation to combine is similar to motivation in claim 1.

Claim 13 is rejected for similar reasons to claim 1 in addition to **mobile carrier and carrier body** *(see Wang fig 1. [0010] electric motorcycle).*

Claim 14 is rejected for similar reasons to claim 2.

2. Claim 7 is rejected under 35 U.S.C. 103 as being unpatentable over Wang US 20190248439 in view of Panchawagh et al. US 20200160018 and further in view of Liu et al. US 20200380240.

Consider claim 7. Wang as modified by Panchawagh disclose **the dashboard device according to claim 1, comprising a cushion material,** but do not disclose **wherein the ultrasonic fingerprint identification assembly is located between the functional interface layer and the cushion material, and a gap is between the ultrasonic fingerprint identification assembly and the cushion material.**

Liu however discloses however discloses **wherein the ultrasonic fingerprint identification assembly is located between the functional interface layer and the cushion material** *fig 3 ultrasonic sensor 230 is between foam layer 220 and display 210.***, and a gap is between the ultrasonic fingerprint identification assembly and the cushion material** *fig 3 see gap.*

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the dashboard device of Wang as modified by Panchawagh to include **wherein the ultrasonic fingerprint identification assembly is located between the functional interface layer and the cushion material, and a gap is between the ultrasonic fingerprint identification assembly and the cushion material** as taught by Liu, to protect the sensor from compression when contact is made with display screen [0145].

3. Claims 10 and 18 are rejected under 35 U.S.C. 103 as being unpatentable over Wang US 20190248439 in view of Panchawagh et al. US 20200160018 and further in view of Raynor et al. US 20190251403.

Consider claim 10. Wang as modified by Panchawagh disclose **the dashboard device according to claim 9,** but do not explicitly disclose **wherein a light transmittance of the light-transmitting protection plate is greater than 94%.**

Raynor however discloses however discloses **wherein a light transmittance of the light-transmitting protection plate is greater than 94%.** *[0139] front part of housing is made of glass and is transparent e.g. has transmittance greater than or equal to 90%.*

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the dashboard device of Wang as modified by Panchawagh to include wherein a light transmittance of the light-transmitting protection plate is greater than 94% as taught by Raynor, to improve impact resistance [0139].

Claim 18 is rejected for similar reasons to claims 9-10.

4. Claims 11 and 19 are rejected under 35 U.S.C. 103 as being unpatentable over Wang US 20190248439 in view of Panchawagh et al. US 20200160018 and further in view of Liu et al. US 20210295004 hereinafter "Liu2".

Consider claim 11. Wang as modified by Panchawagh disclose **the dashboard device according to claim 1,**but do not disclose **wherein a driving voltage of the ultrasonic fingerprint identification assembly is greater than 1000 millivolts.**

Liu2 however discloses **wherein a driving voltage of the ultrasonic fingerprint identification assembly is greater than 1000 millivolts.** *[0062] ultrasonic signal in response to alternative*

current [0064] 200V or 50V 100V. the second voltage can be greater than or equal to 1V and less than or equal to 10V.

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the dashboard device of Wang as modified by Panchawagh to include **wherein a driving voltage of the ultrasonic fingerprint identification assembly is greater than 1000 millivolts** as taught by Liu2, so that the ultrasonic signal can reach a users' finger through the respective film layers [0062].

Claim 19 is rejected for similar reasons to claim 11.

5. Claim 12 is rejected under 35 U.S.C. 103 as being unpatentable over Wang US 20190248439 in view of Panchawagh et al. US 20200160018 and further in view of Kim et al. US 20210004117.

Consider claim 12. Wang as modified by Panchawagh disclose **the dashboard device according to claim 1, wherein the ultrasonic fingerprint identification assembly comprises a fingerprint sensor** *Panchawagh fig 1b*, **a controller** *Panchawagh fig 1b 115*, **a connector** [0052], **and** a **flexible printed circuits** *fig 2 FPC 250 [0052],*

Motivation to combine is similar to motivation in claim 1.

Wang as modified by Panchawagh however does not disclose **two flexible printed circuits one of the two flexible printed circuits is connected between the fingerprint sensor and the controller, and the other of the two flexible printed circuits is connected between the connector and the controller.**

Kim however discloses **two flexible printed circuits (***first flexible circuit 1171 and second flexible printed circuit board 1121***) one of the two flexible printed circuits is connected between the**

fingerprint sensor and the controller, and the other of the two flexible printed circuits is connected between the connector and the controller Fig. 11A-B first flexible printed circuit 1171 connected between and connector of mainboard and second flexible printed circuit 1121 is connected processor and connector of main circuit board.

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the dashboard device of Wang as modified by Panchawagh to include **one of the two flexible printed circuits is connected between the fingerprint sensor and the controller, and the other of the two flexible printed circuits is connected between the connector and the controller** as taught by Kim, so as to transmit to at least one processor, power and a signal for controlling the display panel 1120 [0119].

6. Claim 20 is rejected under 35 U.S.C. 103 as being unpatentable over Wang US 20190248439 in view of Lee et al. US 20150324570.

Consider claim 20. Wang discloses **an authentication** *[0135] fingerprint authentication* **method for a mobile carrier** *fig 1 electric scooter or electric motorcycle [0010],* **comprising:**

using a fingerprint identification assembly of the mobile carrier to identify a fingerprint of a user to obtain an identification result *[0135] biometric reader used to acquire fingerprint. Fingerprint provides an authentication of the rider*, **executing a predetermined procedure corresponding to the at least one fingerprint data.** *[0135] The fingerprint provides an authentication of the rider before the activation of the electric scooter 100. If fingerprint is not matched then electric scooter will not be activated.*

Wang does not disclose **establishing a fingerprint database, wherein the fingerprint database comprises at least one fingerprint data; determining whether the fingerprint of the user matches the at least one fingerprint data according to the identification result;**

Lee however discloses **establishing a fingerprint database, wherein the fingerprint database comprises at least one fingerprint data** *[0089]]compare fingerprint input from sensor to fingerprint information stored in at lease one of memory of the server,* **determining whether the fingerprint of the user matches the at least one fingerprint data according to the identification result fig 14 fingerprint recognition** *[0091 determine whether fingerprint matches* **; and if the fingerprint of the user matches the at least one fingerprint data, executing a predetermined procedure corresponding to the at least one fingerprint data** *[0092] perform a predetermined function corresponding to a specific fingerprint of a specific user based on the acquired fingerprint (or fingerprint information). For example, each recognition of each fingerprint may run different applications, may perform an ID and password input function for different sites in place of a user, or may perform a finger-specific function such as the volume up/down function or the TV channel up/down function.*

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the dashboard device of Wang to include **establishing a fingerprint database, wherein the fingerprint database comprises at least one fingerprint data; determining whether the fingerprint of the user matches the at least one fingerprint data according to the identification result and if the fingerprint of the user matches the at least one fingerprint data, executing a predetermined procedure corresponding to the at least one fingerprint data**, as taught by Lee, to enable different function to execute based on different fingerprint information [0092].

IV. CONCLUSION

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Van Ostrand et al. US 20230086442 discloses an ultrasonic sensor for fingerprint scanning.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to IBRAHIM A KHAN whose telephone number is (571)270-7998. The examiner can normally be reached on 10am-6pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, LunYi Lao can be reached on 5712727671. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

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